

Multiobjective Deep Reinforcement Learning Driven Collaborative Optimization of TSV-Based Microchannel and PDN for 3-D ICs

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Abstract—This study introduces a multiobjective deep reinforcement learning (MODRL) framework for the concurrent thermal-hydraulic optimization of the through-silicon-via (TSV) microchannel heat sink (MCHS) embedded in 3-D integrated circuits (3-D ICs) power delivery network (PDN). By exploiting the inherent structural synergy between TSVs and pin-fin MCHS, the proposed method enhances thermal management in high-density 3-D ICs. The framework integrates deep reinforcement learning (RL) with multiobjective optimization and computational fluid dynamics (CFD) simulations, enabling an efficient exploration of the high-dimensional design space to resolve tradeoffs between thermal efficacy and fluidic resistance. Relative to baseline, the optimized design achieves a reduction in maximum chip temperature of up to 3.3% while concurrently lowering the overall pressure drop by 17.2%. Impedance analysis further validates the design's superiority, showing that the optimized TSV geometry effectively suppresses high-frequency peak impedance. Compared with standard deep reinforcement learning (SDRL) and genetic algorithm (GA), MODRL converges faster by 57.1% and 62.5%, respectively, showing stronger convergence. These results highlight the advantages of the MODRL intelligent optimization framework in design speed and its great potential in driving the development of next-generation 3-D integrated circuits, especially in applications requiring high power density and high reliability.

Index Terms—Interposer microchannel heat sink (MCHS), machine learning (ML), multiobjective deep reinforcement learning (MODRL), through-silicon vias (TSVs).

I. INTRODUCTION

AS TRANSISTOR scaling nears the physical limits of Moore's law [1], 3-D integrated circuits (3-D IC) have emerged as a promising solution [2]. By vertically stacking dies through through-silicon vias (TSVs), 3-D ICs reduce interconnect delay, increase I/O density, and lower power

consumption compared to planar architectures [3]. However, the increased power density in stacked structures leads to localized hotspots, degrading electrical performance and inducing thermomechanical stress that can cause interface delamination and electromigration failures [4].

To address these thermal challenges, microchannel heat sinks (MCHSs) have been introduced. Their compact size allows integration into interposers, and their high surface-to-volume ratio enables efficient heat removal. Since Tuckerman's seminal work [5], numerous designs have enhanced cooling performance. Feng et al. [6] proposed a hybrid pin-fin MCHS (HPFMCHS) with region-specific structures, while Back et al. [7] developed a hierarchical manifold MCHS that cooled chips with volumetric heat densities up to 2870 W/cm³. Given the structural similarity between TSVs in 3-D ICs and MCHS pin-fins, TSVs can be integrated with heat sink fins [8]. Feng et al. [9] proposed a design of jet impingement cooling that combines TSV improving chip temperature uniformity by 12.21%. Bakir et al. [10] fabricated TSV-based interconnects with embedded microchannels. Oh et al. [11] subsequently developed high aspect ratio TSVs within micro pin-fin structures. Zhang et al. [12] engineered an embedded TSV pin-fin heat sink and achieved an optimal tradeoff. Yan et al. [13] proposed a hybrid 3-D IC cooling module that integrates a microfluidic network and TSVs and co-optimized microchannel geometries and TSV.

In 3-D ICs, integrating TSVs with pin-fin heat sinks requires careful geometric and structural optimization to enhance heat dissipation, while the power distribution network (PDN) must ensure stable voltage delivery to functional blocks [14]. Stacked chip PDNs are interconnected via TSV arrays, where higher TSV densities improve electrical transmission but also increase interlayer thermal resistance. This tradeoff between electrical performance and thermal management necessitates a layout strategy that simultaneously addresses heat dissipation and TSV topology [15]. Empirical co-optimization is often time-consuming and suboptimal. With advances in machine learning (ML), deep reinforcement learning (DRL) is well suited for co-optimizing the power distribution network in 3-D ICs with TSV pin-fin structures, due to its strength in solving high-dimensional, dynamic problems through reward-driven exploration [16]. Guan et al. [17] proposed a novel TSV allocation method based on multiagent DRL, which outperformed the traditional method in terms of line length

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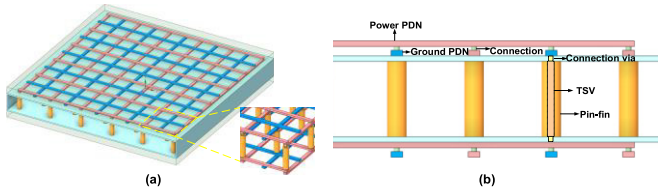


Fig. 1. (a) Double-layer 3-D IC PDN with integrated pin-fin microchannels. (b) Cross section of pin-fin microchannel and PDN structure.

and temperature. Kim et al. [18] proposed an improved DRL-optimized TSV array method to minimize far-end crosstalk by 3 dB. Li et al. [19] proposed a physics-assisted DRL method to optimize the signal integrity/power integrity (SI/PI) of TSV arrays, achieving faster convergence and better SI/PI quality.

To enhance the multiobjective optimization capabilities of traditional DRL algorithms, this study proposes a novel multiobjective DRL (MODRL) framework for the rapid and coordinated optimization of thermal-hydraulic performance in through-silicon-via (TSV) MCHSs embedded within the power delivery network (PDN) of 3-D integrated circuits (3-D ICs). The principles of the MODRL algorithm and its reward design are first elaborated to explain its multiobjective optimization mechanism. A sensitivity analysis of key structural parameters is then conducted to validate the effectiveness of the selected optimization variables. The MODRL framework is subsequently applied to jointly optimize the shape and spatial distribution of TSV pin-fins, achieving significant synergistic improvements in both thermal and hydraulic performance. Computational fluid dynamics (CFD) simulations of the optimized structure quantify these improvements. Electrical performance tests further confirm that the optimized TSVs maintain excellent electrical integrity. Finally, the convergence efficiency of the proposed MODRL approach is compared with traditional optimization methods, such as the genetic algorithm (GA) and standard DRL, demonstrating faster convergence and superior solution quality.

II. PROPOSED MODULATOR ARCHITECTURE

Fig. 1(a) shows the double-layer PDN structure of the 3-D IC used in this study. Since this study mainly analyzes the thermal design of PDN, the specific structure of PDN is not emphasized. Therefore, the PDN model proposed by Li et al. [20] is adopted. PDN is equivalent to a network structure with two layers of rectangular metal wires superimposed. The red metal is the power PDN, and the blue is the ground PDN. A PDN unit is shown in the enlarged structure diagram of Fig. 1(a). Each layer of PDN consists of 5×5 units, and a total of two layers of PDN are connected by TSV. The length (L_{PDN}), width (W_{PDN}), and height (H_{PDN}) of the metal wire are 2030, 30, and 10 μm , respectively.

Since full-scale simulation consumes computing resources and is not conducive to overall optimization efficiency, a simplified model is performed. This study is only concerned with the combination of two PDN units and eight TSV pin-fin microchannels and uses symmetric boundary conditions to expand in subsequent simulations to achieve indirect full model research. The pitch between the power line and the power line is 400 μm , while the pitch between the power line and the ground line is 200 μm . The spacing between each layer

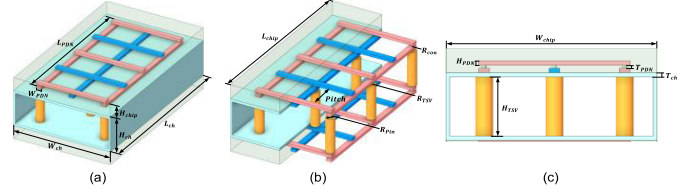


Fig. 2. (a) TSV pin-fin microchannel containing double unit PDNs. (b) Cross section of pin-fin microchannel and PDN structure. (c) Front view of a TSV pin-fin microchannel containing two units of PDNs.

of the network (T_{PDN}) in a layer of PDN is 10 μm , and the middle is connected by a cylindrical copper (R_{con}) contact with a radius of 10 μm . The length (L_{ch}), width (W_{ch}), and height (H_{ch}) of the microchannel are set to 2030, 630, and 228 μm , respectively. The thickness of the microchannel (T_{ch}) is set as 14 μm , so the PDN and TSV are connected by a copper-filled through-hole with a height of 14 μm . The simplified model of chips is with a length (L_{chip}), width (W_{chip}), and height (H_{chip}) of 2030, 630, and 20 μm , respectively. The height of the TSV (H_{TSV}) is set as 200 μm and the radius (R_{TSV}) is set as 15 μm . The height of the pin-fin is the same as TSVs and the radius (R_{pin}) is set as 25 μm . Fig. 2 shows the overall structure, cross-sectional view, and front view of the simplified model used in this study.

III. BOUNDARY CONDITIONS AND MATERIAL PROPERTIES

Heat transfer and fluid flow simulations are performed in COMSOL 6.0 Multiphysics software, with temperatures in kelvin (K) and pressures in kilopascals (kPa). The flow model is 3-D, steady, and laminar, with no phase change; gravity and thermal radiation are neglected. The heat sink walls are considered adiabatic, and symmetry and periodic boundary conditions are applied to the microchannel sides. Since the focus of this study is to optimize the heat sink structure by a novel algorithm to achieve better heat dissipation efficiency rather than the rate of the microfluidic pump, the boundary conditions such as the microfluidic flow rate are kept unchanged as much as possible [8], [20], [13]. The mass flow rate of inlet is uniform at $2.1546 \times 10^{-4} \text{ kg} \cdot \text{s}^{-1}$, and the inlet temperature is 293 K. A static pressure boundary condition of 0 kPa is applied at the outlet. According to [21], many desktop central processing units (CPUs) from Intel and AMD allow users to run them at higher frequencies for enhanced performance. The thermal design power (TDP) at base frequency may exceed 200 W, and the actual power consumption after overclocking may significantly exceed the rated TDP. Going forward, thermal management challenges are expected to focus more on managing heat flux (W/cm^2) rather than total TDP (W). Even with increasing TDP, acceptable heat flux levels can still be maintained through continuous refinement of 3-D IC design. Therefore, the power density for both chips is set as 300 W/cm^2 . No slip conditions are applied to all solid-liquid interfaces.

Copper is used for PDN, connecting vias, and TSV. Silicon is used for microchannels and pin-fins. Deionized water is used as the fluid, and its temperature-dependent thermophysical properties are modeled using (1)–(4). The material properties were obtained from the study by Luo et al. [22] and

TABLE I
THERMOPHYSICAL PROPERTIES OF MATERIALS

Variable	Density	Specific heat capacity	Thermal conductivity	Dynamic viscosity
Symbol	ρ	c_p	k	μ
Unit	$\text{kg}\cdot\text{m}^{-3}$	$\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	$\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	$\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}$
Silicon	2329	712	148	-
Copper	8960	100	398	-
Water	(1)	(2)	(3)	(4)

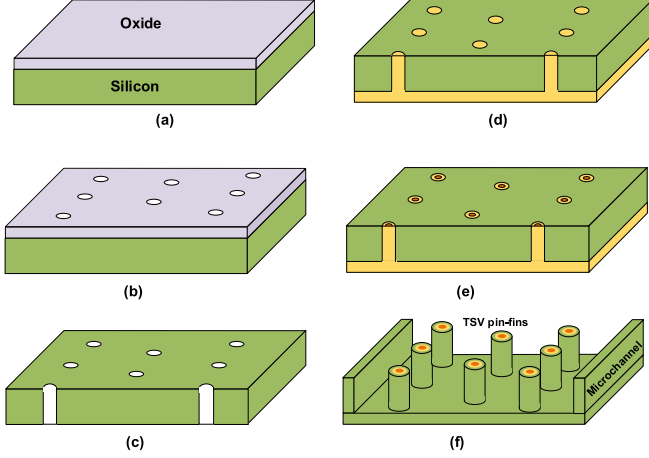


Fig. 3. TSV manufacturing process with embedded microchannel pin-fins. (a) Oxide deposition. (b) Oxide etch. (c) Bosch process (silicon etch). (d) Copper electroplating. (e) Metal deposition and CMP. (f) Fabricate structures using Bosch.

Chung et al. [8], and the complete material properties are listed in Table I

$$\rho_w(T) = 999.9 + 9.561 \times 10^{-2}T - 1.013 \times 10^{-2}T^2 \quad (1)$$

$$C_{p,w}(T) = 4217 - 3.452 T + 1.155 \times 10^{-1}T^2 \quad (2)$$

$$k_w(T) = 5.698 \times 10^{-1} + 1.772 \times 10^{-3}T - 4.870 \times 10^{-6}T^2 \quad (3)$$

$$\mu_w(T) = 1.750 \times 10^{-3} - 5.558 \times 10^{-5}T + 1.172 \times 10^{-6}T^2. \quad (4)$$

IV. FABRICATION OF EMBEDDED TSVs' MICROCHANNEL PIN-FIN HEAT SINK

This section describes the fabrication process for TSV-embedded microchannels and micro pin-fin heat sinks. The feasibility of the process was verified in a study by Oh et al. [23]. Fig. 3 illustrates the fabrication flow of TSVs within the microfluidic heat sinks.

As shown in Fig. 3(a), the process begins with the deposition of a silicon dioxide layer on the surface of a silicon wafer. Next, a thin chrome film is used as a mask to perform anisotropic dry etching of the oxide, as illustrated in Fig. 3(b). Using the patterned silicon dioxide as the etching mask, the silicon wafer is etched with a high-aspect-ratio Bosch process to form deep vias, as shown in Fig. 3(c).

After silicon etching and removal of the remaining silicon dioxide, the wafer is placed in a Tystar furnace for oxide growth to electrically isolate the copper-filled TSVs from the silicon substrate. As shown in Fig. 3(d), a two-step copper electroplating process is then carried out to fill the vias.

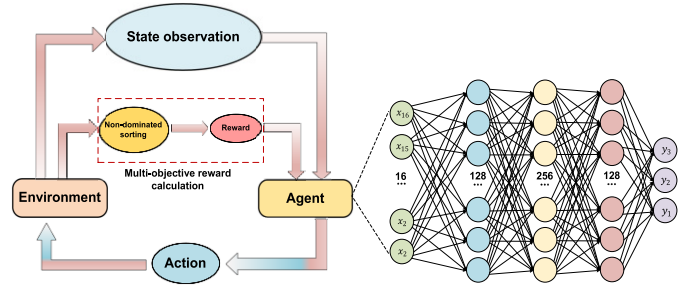


Fig. 4. DRL workflow with CFD environment.

First, a metal seed layer is deposited on the backside of the wafer, and the vias are sealed using dc copper electroplating from the backside. Second, a pulsed electroplating process is employed from the front side of the wafer, enabling bottom-up copper filling of the TSVs. Upon completion of electroplating, chemical mechanical polishing (CMP) is performed from both sides of the wafer to remove excess copper. Following CMP, as illustrated in Fig. 3(e), titanium/copper/gold multilayer metal pads are deposited on the front side of the wafer to facilitate probing. Gold passivation is subsequently applied to the pads to prevent oxidation.

In the final step, shown in Fig. 3(f), another Bosch process is performed from the backside of the wafer to fabricate the microchannel and micro pin-fins structures. In this step, cylindrical silicon posts are formed around the TSVs for micro pin-fins structures, while rectangular trenches are created for microchannels.

V. MULTIOBJECTIVE DRL

Traditional reinforcement learning (RL) techniques struggle with high-dimensional, continuous state and action spaces, limiting their applicability to complex tasks such as heat sink design. Multiobjective DRL (MODRL), by contrast, employs deep neural networks to learn optimal policies in large-scale environments, overcoming these limitations. Among DRL algorithms, proximal policy optimization (PPO) stands out for its stability, efficient convergence, and resistance to overfitting in both discrete and continuous action spaces. PPO alternates between data collection and policy updates via stochastic gradient ascent and constrains policy updates by clipping probability ratios. In this study, PPO clipping is used to optimize pin-fin arrangement and pin-fin distributions. However, due to the difficulty of DRL in multiobjective optimization, in order to improve the performance of DRL in such optimization, this study adds a nondominated selection algorithm and proposes an MODRL. By calculating the dominance relationship between individuals to enrich the reward setting, DRL can optimize multiple objectives at the same time.

The workflow of the MODRL model used in this study is illustrated in Fig. 4. In this framework, the agent determines the subsequent actions based on the rewards and observations received from the CFD simulation environment. Four types of actions are defined to guide the optimization process. The first action involves modifying the TSV height (HTSV) parameter, where a new HTSV value is randomly selected within the range of 100–200 μm and uniformly applied to all TSVs. The second action adjusts the pin radius parameters (RPs) by randomly selecting several pins and assigning them

new random values within the range of 10–100 μm , thereby affecting both the thermal conduction efficiency and fluidic resistance. The third action configures the connection states between TSVs and their corresponding pin-fins, essentially controlling which pins participate in thermal conduction, and thus enabling flexible adjustment of the overall TSV and pin-fin distribution. It is essential to ensure that at least one power TSV and one ground TSV remain active to maintain the fundamental two-layer PDN interconnection. The fourth action introduces hybrid modifications, where a random combination of the aforementioned actions is performed, allowing the agent to explore multiple dimensions of the parameter space simultaneously and enhancing its ability to avoid local optima during the learning process. The reward function is defined by the following equations:

Objective improvement

$$= \frac{T_{1,\text{now}} - T_{1,\text{pre}}}{T_{1,\text{pre}}} + \frac{T_{2,\text{now}} - T_{2,\text{pre}}}{T_{2,\text{pre}}} + \frac{\Delta p_{\text{now}} - \Delta p_{\text{pre}}}{\Delta p_{\text{pre}}} \quad (5)$$

Pareto reward

$$= \frac{\text{number of dominance} - \text{number of dominated}}{\text{number of historical solutions}} \quad (6)$$

Weight reward

$$= w_1 (1 - T_{1,\text{norm}}) + w_2 (1 - T_{2,\text{norm}}) + w_3 (1 - T_{3,\text{norm}}) \quad (7)$$

Total reward

$$= \text{Objective improvement} + \text{Pareto reward} + \text{Weight reward} \quad (8)$$

where $T_{1,\text{now}}$, $T_{2,\text{now}}$, and Δp_{now} denote the current maximal temperatures of chip1, chip2, and the pressure drop of the microchannel, respectively. $T_{1,\text{pre}}$, $T_{2,\text{pre}}$, and Δp_{pre} represent the previous iterations' values, while w represents the importance of temperatures and pressure drop in the reward. In thermal management system design, the most critical design principle is to maintain chip temperature within a safe operating range to ensure overall system reliability and stability. Only after this primary thermal performance requirement is met will secondary optimization objectives—minimizing fluid pressure drop to reduce pumping power, improve flow distribution, and enhance chip temperature uniformity—be considered. This hierarchy of design priorities dictates that when constructing a multiobjective optimization model, temperature improvement must be given a significantly higher reward than pressure drop improvement. Therefore, (w_1, w_2, w_3) is set as (0.4, 0.4, 0.2).

As shown in Fig. 4, the agent consists of three fully connected hidden layers. The hyperparameters of the PPO model are detailed in Table II. The rectified linear unit (ReLU) was adopted as the activation function due to its computational simplicity, fast convergence, and effectiveness in mitigating the vanishing gradient problem. Compared with Sigmoid and Tanh, which suffer from saturated gradients and higher computational cost, ReLU enables deeper networks and better scalability. For optimization, the adaptive moment estimation (Adam) algorithm was used, which adaptively adjusts learning rates based on the first and second moments of the gradients. Unlike traditional stochastic gradient descent (SGD), which requires careful tuning and converges slowly, or RMSprop,

TABLE II
PPO HYPERPARAMETERS EMPLOYED IN THIS STUDY

Hyperparameter	Value
Activation function hidden layers	ReLU
Optimizer	Adam
Hidden layers	3
Nodes hidden layers	128,256,128
Batch size (N)	64
Learning rate / Clipping ratio	0.00001 / 0.2
GAE parameter / Discount factor	0.95 / 0.99

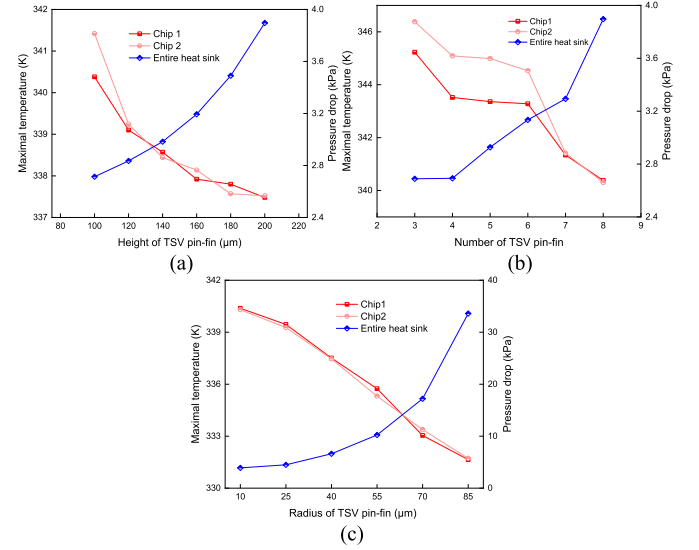


Fig. 5. Parameter sensitivity analysis for (a) TSV pin-fin height, (b) number of TSV pin-fin, and (c) TSV pin-fin radius.

which lacks momentum, Adam combines both techniques to achieve faster and more stable convergence. This combination of ReLU and Adam provides an efficient and robust foundation for training MODRL models. A total of 18 000 episodes were run. The CPU used in this article is an AMD 7995WX with 128 GB of memory and an NVIDIA A800 GPU. The total optimization time is 200 h.

VI. RESULTS AND DISCUSSION

A. Parameter Sensitivity Analysis

Figs. 4(b) and (c) and 5(a) illustrate the effects of TSV pin-fin height, number, and radius on the maximal temperatures of chip1 and chip2, as well as the overall pressure drop across the heat sink.

In Fig. 5(a), it is observed that increasing the height of the TSV pin-fins leads to a gradual decrease in the maximal temperatures of both chips. This improvement is attributed to the enlarged heat exchange surface area between the pin-fins and the coolant, which enhances thermal dissipation efficiency. However, when the height reaches 160 μm , the temperature reduction effect begins to plateau. Between 160 and 200 μm , the temperature differences between adjacent height values are only 0.095% and 0.16%, respectively. The thermal benefit gradually diminishes as the pin-fin height continues to increase, while the increased obstruction area makes the resistance to fluid flow to significantly increase, leading to an increase in pressure drop across the heat sink. Because the pin-fin structure encloses the TSV, changes in

pin-fin height also alter the inlet height of the TSV pin-fin layer. At a constant mass flow rate, this reduces fluid velocity, further amplifying the pin-fin's impact on pressure drop. Ultimately, this increases flow resistance and leads to an increase in pressure drop of up to 11.64%, impairing overall cooling performance.

Fig. 5(b) presents the trends in temperature and pressure drop as the number of TSV pin-fins varies. The spatial distribution of the pin-fins is randomized for each configuration, with the constraint that the total number ranges from 3 to 8, and at least one power TSV and one ground TSV are maintained. As the number of pin-fins increases, the temperature and pressure drop exhibit nonlinear behaviors. A substantial temperature reduction is observed when increasing the number from 3 to 4, primarily due to the expanded heat exchange area. At this stage, the pressure drop remains relatively unchanged, suggesting almost constant flow obstruction. However, when the number increases from 4 to 6, the temperature reduction becomes marginal—only 0.07% and 0.16% for the two chips—while the pressure drop increases by 16.40%. This indicates that the increased thermal benefit is offset by the reduced heat transfer efficiency caused by the increased flow resistance, leading to reduced cooling efficiency. When the number exceeds 7, although the pressure drop continues to rise, the maximal temperatures decrease more significantly, implying that the positive effect of enhanced heat transfer area outweighs the negative impact of reduced coolant velocity.

Fig. 5(c) shows the influence of TSV pin-fin radius on the maximal temperature and pressure drop. As the radius increases, the heat transfer area grows, resulting in a gradual decline in maximal temperature. However, the larger pin-fin radius also introduces greater obstruction to fluid flow, which causes a substantial increase in the pressure drop.

In summary, the TSV pin-fin height, number, and radius significantly influence the thermal and hydraulic performance of the heat sink. These parameters exhibit strong sensitivity and should therefore be considered as key design variables in the optimization of TSV-integrated heat sink structures.

B. DRL Optimization Process Analysis

Fig. 6 shows the evolution curve of rewards during DRL training. The whole process can be divided into four stages. In the initial stage (rounds 0–6000), the reward value fluctuates violently between -160 and 110 , which fully reflects the fact that the agent is still in the random exploration period and the state–action association has not yet been established; at this time, some parameter combinations produce extreme negative rewards due to poor performance, and the violent fluctuations suggest potential optimization space. After entering rounds 6000–9000, there are no more extreme negative rewards, indicating that the agent has found a pin-fin structure that can comprehensively improve the thermal performance of the chip and the pressure drop of the MCHS, but because the reward still shows a relatively violent fluctuation, it can be judged that the agent has not found the optimal structure. In rounds 12000–18000, the algorithm enters the mature stage, and the reward value gradually converges to the high-performance range of 130 – 133 . At this point, the average reward fluctuation between rounds has been controlled within 0.74%,

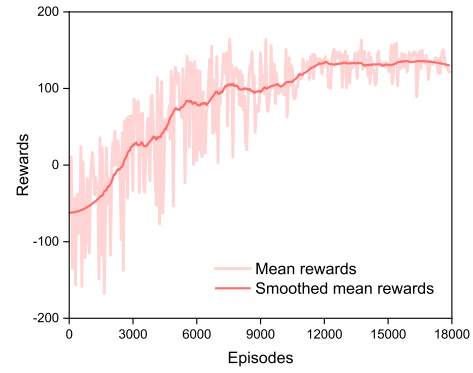


Fig. 6. DRL workflow with CFD environment.

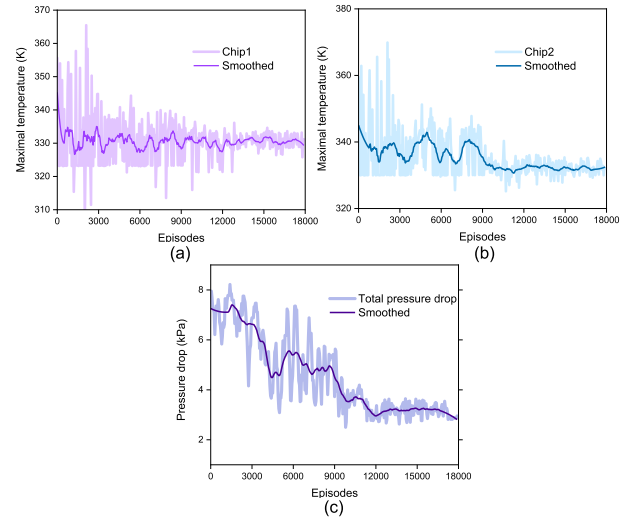


Fig. 7. Contribution of (a) maximal temperature of HP HS, (b) maximal temperature of LP HS, and (c) pressure drop against the episode number.

and the reward distribution density has increased significantly, indicating that the policy network has established a stable state–action mapping, tending to converge, and the entire structure has reached a dynamic balance close to the global optimum. However, around the 17000th round, a few rewards occasionally dropped to 90, indicating that the agent has tried to jump out of the local optimum. The performance has then recovered rapidly, highlighting the key role of the target network in maintaining policy stability. The experience replays and gradually narrowed exploration strategy at this stage has enabled the policy network to continuously strengthen effective behaviors and suppress those parameter combinations that could have led to poor heat transfer or flow performance.

Fig. 7 depicts the evolution of the maximum temperature of chip1, the maximum temperature of chip2, and the overall pressure drop of the microchannel during DRL training with the training rounds. With the iterative optimization of the pin-fin structure, these three indicators all show a change trajectory from large fluctuations to gradual convergence.

In the early stage of training (0–3000 episodes), the maximum temperature of chip1 has fluctuated greatly, with a generally high value, reaching a maximum of 364.85 K, which indicates that the agent has tried a variety of pin-fin layouts in the random exploration stage, and many of these combinations have caused local overheating. After entering the middle stage (3000–12000 episodes), as the effective parameters were

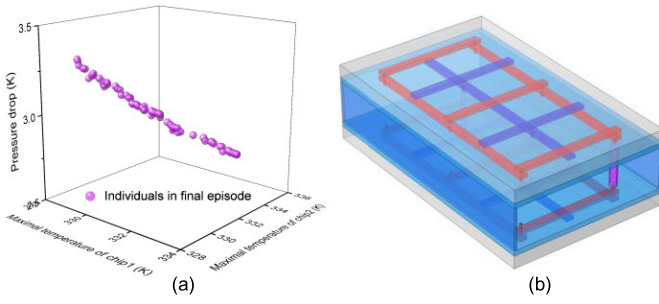


Fig. 8. (a) Pareto front of individuals in the final episode. (b) Optimal model.

screened and repeatedly optimized, the maximum temperature of chip1 has dropped rapidly and began to converge to 330 K. Although some excellent TSV pin-fin distribution structures appear during the period, potentially reducing the temperature to around 320 K, they are eliminated because they degrade the performance of other optimization targets such as temperature and pressure drop. When the training rounds reach more than 12 000, the temperature further stabilizes, with most values concentrated around 330 K, with a slight fluctuation of about 2.4 K on average, indicating that the policy network has mastered the design idea of reducing the temperature in the high power area.

Since chip2 and chip1 are stacked on each other and are the same type of chips with the same power, the maximum temperature change trend is similar. In the first 3000 episodes, some high temperature values appeared in chip2; in the mid-term 3000–12 000 episodes, the experience replay and the gradually converging exploration rate have caused the temperature level to drop significantly and fluctuate in the range of 330–350 K; after the training has entered 12 000 episodes, the maximal temperature of chip2 has further concentrated to about 332 K, and the fluctuation range has narrowed to about 2.6 K, indicating that the intelligent agent has a deep understanding of the heat dissipation law in the low-power area.

The evolution of the overall pressure drop of the system is shown in Fig. 7(c). In the initial exploration phase (0–3000 episodes), the pressure drop data have fluctuated dramatically, indicating that the agent was pursuing cooling effects while also testing the effects of different structures on fluid resistance. As training progressed (3000–12 000 episodes), the pressure drop shows a clear downward trend, but with still large fluctuations between 2 and 8 kPa. When the number of training episodes exceeds 12 000, most of the pressure drop values are stabilized at about 3 kPa, and the average fluctuation between rounds is controlled at around 0.2 kPa. This shows that the agent successfully minimize flow resistance while maintaining heat dissipation performance.

C. Comparison of Optimal Structural Performance

Since this study uses nondominated sorting as part of the MODRL reward, the optimal population generated in the end will be concentrated on the Pareto front. Fig. 8(a) shows the Pareto front of the optimal population. Since the temperature is of higher importance than other properties for chip operation stability, the design with the lowest peak temperature is considered the optimal solution. The corresponding structural schematic is shown in Fig. 8(b). It can be found that the

TABLE III
OTHER SPECIFIC PARAMETERS EMPLOYED IN THIS STUDY

Parameter	Value (μm)
H_{TSV}	106.63
$R_{\text{TSV,POWER}}$	10.89
$R_{\text{TSV,GROUND}}$	10.58

TABLE IV
COMPARISON OF TEMPERATURE AND PRESSURE DROP BETWEEN THE INITIAL AND OPTIMAL MODEL

Variable	T_{max} of chip1	T_{max} of chip2	Pressure drop
Unit	K	K	kPa
Original	340.12	341.42	3.89
MODRL	329.24	330.08	3.22
optimal	(3.2% $\uparrow$)	(3.3% $\uparrow$)	(17.2% $\uparrow$)

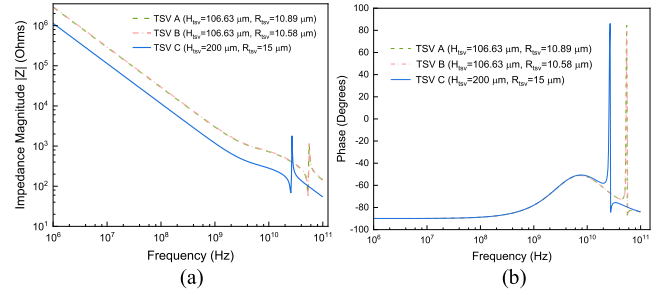


Fig. 9. Comparison of (a) total impedance and (b) phase of the optimal TSV design and the initial design at different frequencies.

final structure has two TSVs, the TSV height (H_{TSV}) is unified to 106.63 μm , and the radii of the power and the ground TSV pin-fins ($R_{\text{TSV,POWER}}$, $R_{\text{TSV,GROUND}}$) are (10.89, 10.85) μm . Table III lists the specific parameters. Table IV compares the performance of the optimal design with the initial structure. The results show that compared with the initial design, the design with integrated TSV pin-fin microchannel after DRL optimization reduces the peak temperature of chips by 3.2% and 3.3%, while the overall pressure drop of the microchannel is reduced by 17.2%.

D. Electrical Performance Verification of the Optimal Design

The calculation of TSV voltage drop conforms to Ohm's law. Therefore, for a given current, impedance affects the voltage drop. Different TSV sizes affect the impedance of the TSV itself, as well as the parasitic impedance surrounding the TSV. The TSV equivalent circuit method proposed by Chung et al. [8] and Kim et al. [24] is used to calculate the impedance of the TSV and its parasitic coupling. Fig. 9(a) and (b) shows the diagrams of the impedance and phase corresponding to TSVs, where TSV A, TSV B, and TSV C represent the initial TSV design and the optimized TSV, respectively.

In the low-frequency dominant region (1 MHz–1 GHz), the impedance characteristics of the three different-sized TSVs are governed by the capacitance effect. This is evidenced by a phase angle that remains consistently close to -90° , indicating purely capacitive behavior. At the starting frequency of 1 MHz, the impedances of TSVs A–C are 2.88, 2.96, and 1.14 M Ω , respectively. The similar impedance values of TSVs A and B show that fine-tuning the radius from 10.89 to 10.58 μm has a negligible impact on low-frequency

parasitic capacitance. However, the impedance of TSV C is only about 39% of that of A and B. This significant difference is because parasitic capacitance is proportional to the conductor's surface area. With a much larger height and radius, TSV C has a substantially increased effective sidewall area. This, in turn, increases the parasitic capacitance between it and the silicon substrate and coolant. The larger capacitance results in a lower capacitive impedance. While TSV C offers a lower impedance path to ground, making it theoretically better for filtering low-frequency noise, this advantage has limited practical significance as the primary challenge in modern chips is high-frequency noise.

The mid-frequency band (1–10 GHz) represents a transitional region for inductance effects. As the frequency enters the GHz range, the series inductance of the TSV itself becomes significant, and the total impedance is no longer determined solely by capacitance, but rather by the interaction of the inductance and capacitance effects. It can be seen that the downward trend in the total impedance is slowing down. This is because the increasing effect of the inductive component of the impedance, which is proportional to frequency, begins to offset the decreasing effect of the capacitive component, which is inversely proportional to frequency.

The high-frequency band (>10 GHz) is the resonance-dominated region and the most critical area for determining the upper performance limit of a TSV. This region is characterized by sharp impedance spikes caused by parallel resonance with the phase angle changing from negative (capacitive) to 0° (purely resistive) to positive (inductive) near the resonance frequency. TSV C exhibits an extremely sharp resonance peak of $1780\ \Omega$ at 26.84 GHz. Conversely, the resonance peaks for TSVs A and B occur at 56.17 GHz, with significantly lower peak impedances of approximately 1135 and 1161 Ω , respectively. The peak impedance of C at its resonant point is 1.56 times that of A and 1.53 times that of B. This resonant peak acts as a forbidden zone for signal transmission, where any signal with frequency components falling within it will be severely corrupted. The designs of A and B push this forbidden zone far beyond the frequency range of current mainstream high-speed signals, ensuring reliable signal transmission. In contrast, the bandwidth limitations of C make it unsuitable for these applications. The high resonance peak of TSV C may have a significant impact on the performance of the PDN. If the switching harmonics of a chip module, such as a PLL or high-speed I/O, were to fall at 26.8 GHz, the resulting 1780- Ω impedance would generate catastrophic voltage noise, highly likely causing system failure. TSVs A and B not only shift this risk to a safer higher frequency but also reduce the peak impedance by approximately 1.5 times.

In summary, the optimized design demonstrates lower peak impedance at high frequencies, which provides superior high-frequency performance compared to the original design.

E. Algorithm Performance Verification

To validate the effectiveness of the proposed MODRL, two baseline algorithms were selected for comparison: the GA and standard DRL (SDRL).

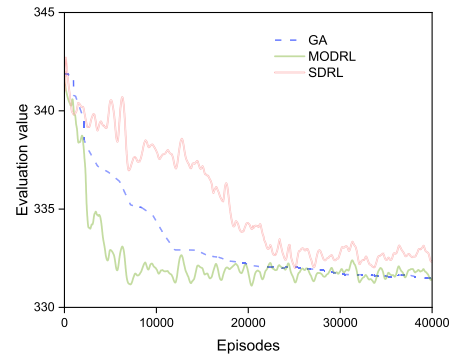


Fig. 10. Comparison of convergence speed and optimization performance of GA, MODRL, and TDRL.

The GA, inspired by biological evolution, is a stochastic global optimization method that simulates the processes of selection, crossover, and mutation. Starting from a randomly initialized population, GA iteratively evolves candidate solutions through these mechanisms, progressively exploring the search space and converging toward optimal solutions. As a classical and widely used multiobjective optimization approach, GA was chosen as a benchmark. In addition, SDRL was included to demonstrate the advantages of MODRL over conventional RL methods that lack explicit multiobjective handling.

Fig. 10 compares the convergence performance of GA, SDRL, and MODRL. Since GA typically uses fitness value as an evaluation metric, while DRL-based methods rely on reward signals, a unified metric was adopted to ensure fair comparison. In this study, the sum of average maximal temperature and pressure drop was used as the evaluation criterion, capturing both thermal and hydraulic performance of the heat sink. As shown in the figure, MODRL achieves a rapid reduction in the evaluation metric within the first 10,000 episodes, decreasing by 2.91% from the initial value, compared to only 2.2% for GA and 1.15% for SDRL. MODRL reaches near convergence by 12,000 episodes, with subsequent fluctuations limited to 0.003%, indicating high stability. In comparison, GA and SDRL require approximately 28,000 and 32,000 episodes to converge, whereas MODRL achieves convergence with 57.14% and 62.50% less episodes, respectively, thereby demonstrating significantly stronger convergence efficiency.

The superior convergence speed of MODRL stems from its ability to learn and exploit patterns along the Pareto front. Unlike GA, which relies on random sampling and population sorting, MODRL agents are capable of directly generating high-quality solutions by learning from previous experience. Furthermore, unlike SDRL, which optimizes a single objective per episode, MODRL incorporates Pareto-based reward shaping, enabling simultaneous and balanced optimization of multiple objectives.

These results clearly demonstrate that MODRL outperforms GA and SDRL in both convergence speed and effectiveness for multiobjective optimization tasks.

VII. CONCLUSION

In this study, an MODRL framework was proposed to optimize the thermal and hydraulic performance of a 3-D IC

PDN based on TSV MCHS integrated in a silicon interposer. By leveraging the structural synergy between TSV and pin-fin microchannels, the design framework had addressed the key challenges of thermal management and power integrity in high-density 3-D ICs.

The proposed MODRL method innovatively combined multiobjective optimization with RL, overcoming the limitation of traditional RL algorithms of only effectively optimizing a single objective. The combination of MODRL and CFD simulation enabled to systematically explore the high-dimensional design space and balance conflicting objectives such as thermal performance and fluid resistance. Compared with the unoptimized design, the MODRL optimization framework reduced the maximum temperature of the stacked chips by 3.3% and 3.2% and reduced the overall pressure drop of the MCHS by 17.2%. The optimized TSV pin-fin layout enhanced the fluid flow characteristics and achieved efficient heat dissipation while minimizing the pressure drop. In addition, the impedance of the optimal structure and the initial structure was calculated, which proved that the optimal TSV size has a great advantage in high-frequency peak impedance performance. Finally, the proposed MODRL algorithm is compared with SDRL and GA. The results show that MODRL achieves faster convergence than GA and SDRL, as it requires 57.14% and 62.50% fewer episodes to converge, respectively. This clearly demonstrates a significantly stronger convergence efficiency.

The results highlighted the transformative potential of ML-driven codesign strategies in overcoming the limitations of traditional trial-and-error approaches. At the same time, the proposed MODRL optimization framework has a faster convergence speed and design space than the traditional ML optimization framework. This points to a greater speed advantage. By embedding TSVs as electrical interconnects and thermal management elements, this work established a scalable paradigm for next-generation 3-D ICs, especially in applications that require high power density and high reliability. Future research will focus on extending this framework to multilayer 3-D IC stacks, incorporating dynamic thermoelectric coupling effects, and exploring advanced materials to further improve performance. This approach paves the way for intelligent adaptive thermal management systems in heterogeneous integration and AI-driven electronics.

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